

PARIKSHIT DESAI

gurudesai2005@gmail.com · Pune, India

linkedin.com/in/parikshit-desai-7abb1328a · github.com/Parikshit09-coder · parikshit09-coder.github.io

Information Technology undergraduate at PICT with strong software development skills and experience in full-stack web development, API design, and cloud deployment. Passionate about building scalable applications and exploring emerging technologies like quantum computing.

EDUCATION

Pune Institute of Computer Technology (PICT), Pune
B.Tech in Information Technology

Expected Grad: 2027
CGPA: 9.65

WORK EXPERIENCE

Software Engineering Intern
Citrus Oncology

March 2026 to Present
Remote

- **Real-time AI voice agent:** built using an external **speech-to-text service** and a cloud **call/telephony service** with advanced **interruption handling** and intent classification for natural, low-latency conversations; detected caller barge-ins mid-sentence, halted in-flight LLM generation, and resumed conversations with preserved context.
- Integrated **third-party AI services** (LLMs, speech APIs) into production workflows via scalable **microservices**, following industry-standard security and compliance practices.
- Defined **Protobuf** schemas and set up **gRPC** communication between internal microservices, replacing slower REST calls for high-throughput, low-latency service-to-service messaging.
- **Text-based communication channel:** built on a cloud **SMS service** with custom **number-pool allocation logic** to route messages at scale, surfacing SMS threads directly on the web dashboard for real-time coordination.

Web Developer Intern

July 2025 to March 2026
Remote

TrevasQ (Quantum Computing Research and Development)

- Built **interactive dashboards** and **REST APIs** to visualize and serve research data for the team.
- Built a **document signing platform** end to end (frontend, backend, signing workflow and in-browser signing flow), integrating an existing **post-quantum, hash-based signature library** for key generation and the core signing function.
- Implemented **dual (classical and post-quantum) signature verification** in the platform.
- Optimized **frontend performance** across both applications, reducing load times and improving rendering efficiency for data-heavy dashboards and interactive UI components.

TECHNICAL SKILLS

Languages	C++, Python, JavaScript, TypeScript, HTML, CSS
Frontend	ReactJS, React-Redux/Toolkit, React-Native, Tailwind
Backend	NodeJS, ExpressJS, FastAPI, WebSockets, GraphQL, gRPC
AI/ML	RAG, LangGraph, Speech-to-Text APIs, Aider
Databases	MySQL, MongoDB, Firebase, Redis
DevOps/Cloud	Docker, Kubernetes, AWS (EC2), Argo CD, CI/CD, Nginx, Git, GitHub Actions

PROJECTS

Scale0: AI-Driven Self-Healing Infrastructure

June 2026

Tech: Python, LangGraph, Aider, Next.js, MongoDB, Kubernetes, Docker, FastAPI [GitHub](#)

- Built a closed-loop **GitOps + AIOps self-healing system** that autonomously detects OOMKills/crashes, performs AI root cause analysis, generates fixes, and deploys via Argo CD, eliminating manual remediation.
- Orchestrated a multi-node AI pipeline using **LangGraph** (triage, strategy, execution, and verification stages) with **Claude 3.5 Sonnet** as reasoning manager and **Aider** as the code-editing worker agent.
- Implemented a **DoS Circuit Breaker** to distinguish malicious traffic from bugs; enforced Feasibility Gates with automatic **Jira + email escalation** for human handoff.

MedPass: Complete Health Document Portal

Jan to May 2025

Tech: MERN, MongoDB, Firebase, RAG, Vector DB, Llama LLM, QR Codes [GitHub](#)

- Designed an end-to-end **RAG pipeline** for AI medical-summary generation: OCR/text extraction, chunking, vector embeddings in a **vector database**, semantic retrieval, and **Llama LLM** generation, producing accurate summaries in seconds.
- Built a **RAG-powered chatbot** letting patients/doctors query medical history conversationally, retrieving relevant embedded chunks to ground responses and reduce hallucination.
- Implemented secure **QR-based access control** with time-bound (2-hour) consent sessions for instant, paperwork-free grant/revoke of doctor access, backed by **doctor verification** via the Indian Medical Registry API.
- Architected the backend with **MongoDB** for auth data and **Firebase** for encrypted document storage, aligned with **DISHA** healthcare data-privacy standards.

ACHIEVEMENTS & HACKATHONS

Winner: Xenathon Hackathon 2026 (XENIA26): Built Kubernetes-native traffic system with HPA, Redis caching, Cloudflare CDN, and Jenkins CI/CD to handle concurrent user surges.

2nd Runner-Up: Innovation Software Hackathon 2026, Cummins College (collab. Williams-Sonoma): Scale0 AIOps self-healing pipeline; recognized for GitOps-driven autonomous incident remediation.

MHT-CET: 99.51 percentile | JEE Mains: 96.03 percentile